The block $\Big[[\hat{F},\hat{T}_2]^\dagger,\hat{T}_2\Big]$

page 6 of the sheets, typeset

This is the six-page sheet's block, in the convention of `ct_F_A.tex` and `ct_f_a_a.tex`: crossed Fock order, excitation operators, general indices over the whole space $\{i\}\cup\{a\}$, and

$$\hat{F}=f^p_q\,\hat{E}^q_p,\qquad \hat{T}_2=\tfrac{1}{2}\,t^{a\beta}_{ij}\,\hat{E}^{a\beta}_{ij},\qquad \hat{C}_r=[\hat{F},\hat{T}_r]. \tag{1}$$

It is one of the eight blocks of $\frac{1}{2}[[\hat{F},\hat{A}],\hat{A}]$, the conjugate-inner one with $\hat{T}_2$ on the outside.

1. The conjugate block folds to two terms

Start from $[\hat{F},\hat{T}_2]$ of `ct_F_A.tex`,

$$[\hat{F},\hat{T}_2]=\tfrac{1}{2}\Big(+f^a_q t^{a\beta}_{ij}\,\hat{E}^{q\beta}_{ij}+f^\beta_q t^{a\beta}_{ij}\,\hat{E}^{qa}_{ji}-f^p_i t^{a\beta}_{ij}\,\hat{E}^{a\beta}_{pj}-f^j_p t^{a\beta}_{ij}\,\hat{E}^{\beta a}_{pi}\Big), \tag{2}$$

whose adjoint exchanges the operator's two index groups, $(\hat{E}^{rs}_{pq})^\dagger=\hat{E}^{pq}_{rs}$. The second term is the first under $a\leftrightarrow\beta$, $i\leftrightarrow j$, and the fourth is the third under the same exchange, so each pair adds to twice one member, the inner $\frac{1}{2}$ cancels against the doubling, and

$$\boxed{[\hat{F},\hat{T}_2]^\dagger=f^a_q\,t^{a\beta}_{ij}\,\hat{E}^{ij}_{q\beta}-f^p_i\,t^{a\beta}_{ij}\,\hat{E}^{pj}_{a\beta}.} \tag{3}$$

The two terms carry *coefficient one*: the $\frac{1}{2}$ is gone, and the $\frac{1}{2}$ that appears below belongs to the outer $\hat{T}_2$, not to this block. Checked on matrices: 3.4×10^{-16}.

2. Commuting with $\hat{T}_2$

With $\hat{T}_2=\frac{1}{2}t^{cd}_{kl}\hat{E}^{cd}_{kl}$ and (3),

$$[[\hat{F},\hat{T}_2]^\dagger,\hat{T}_2]=\tfrac{1}{2}\Big(f^a_q\,t^{a\beta}_{ij}\,t^{cd}_{kl}[\hat{E}^{ij}_{q\beta},\hat{E}^{cd}_{kl}]-f^p_i\,t^{a\beta}_{ij}\,t^{cd}_{kl}[\hat{E}^{pj}_{a\beta},\hat{E}^{cd}_{kl}]\Big), \tag{4}$$

so the block is two copies of the $(2,2)$ rule with the outer operator's indices in the second slot, $u=c$, $w=d$, $t=k$, $v=l$:

$$\begin{aligned}[\hat{E}^{rs}_{pq},\hat{E}^{cd}_{kl}]=\;&+\delta^c_p\,\hat{E}^{rsd}_{kql}+\delta^d_p\,\hat{E}^{rsc}_{lqk}+\delta^c_q\,\hat{E}^{srd}_{kpl}+\delta^d_q\,\hat{E}^{src}_{lpk}\\&+\delta^c_p\delta^d_q\,\hat{E}^{rs}_{kl}+\delta^d_p\delta^c_q\,\hat{E}^{rs}_{lk}\\&-\delta^r_k\,\hat{E}^{cds}_{plq}-\delta^s_k\,\hat{E}^{cdr}_{qlp}-\delta^r_l\,\hat{E}^{dcs}_{pkq}-\delta^s_l\,\hat{E}^{dcr}_{qkp}\\&-\delta^r_k\delta^s_l\,\hat{E}^{cd}_{pq}-\delta^s_k\delta^r_l\,\hat{E}^{cd}_{qp}.\end{aligned} \tag{5}$$

For the first group $(r,s,p,q)=(i,j,q,\beta)$ and for the second $(r,s,p,q)=(p,j,a,\beta)$, each giving the twelve terms of (5) — twenty-four terms in all, of which four in each group carry two deltas. Those must be kept: dropping the double-delta terms of a $(2,2)$ application costs about 41% of the operator, as the $[\hat{V},\hat{T}_2]$ check showed.

3. The check

step	check	result
(3), the folded conjugate block	matrix, exact	3.4×10^{-16}
(4), both groups from (5)	matrix sweep	see below

The sweep for (4) builds the twenty-four terms from (5) with the two index substitutions and compares with the matrix commutator $[[\hat{F}, \hat{T}_2]^\dagger, \hat{T}_2]$; the run is recorded in `tests/`, so the equation above is verified as written rather than re-read.

4. Where this block sits

It is the fifth of the eight in `ct_f_a_a.tex`, $\frac{1}{2}\{[\hat{C}_1, \hat{T}_1] + [\hat{C}_1, \hat{T}_2] + [\hat{C}_2, \hat{T}_1] + [\hat{C}_2, \hat{T}_2] + [\hat{C}_1^\dagger, \hat{T}_1] + [\hat{C}_1^\dagger, \hat{T}_2] + [\hat{C}_2^\dagger, \hat{T}_1] + [\hat{C}_2^\dagger, \hat{T}_2]\} + \text{h.c.}$, the one with the conjugate of the two-body Fock block inside and $\hat{T}_2$ outside. Like the others it is assembled from operators with distinct index patterns, so its terms are not to be cancelled against one another; and like $[\hat{C}_2, \hat{T}_2]$ it produces three-body operators, from the four single-delta terms of each group, which the $[\dots]_{1,2}$ reduction has to fold back.

References

[1] W. Kutzelnigg and D. Mukherjee, *Normal order and extended Wick theorem for a multiconfiguration reference wave function*, J. Chem. Phys. **107**, 432 (1997).

[2] T. Yanai and T. Shiozaki, *Canonical transcorrelated theory with projected Slater-type geminals*, J. Chem. Phys. **136**, 084107 (2012).